

Question

- Under anaerobic conditions, a Cu^{2+} containing solution can act as an oxidizing agent to react with pyrite, and the product contains sulfur in two oxidation states: -2 and +6; the oxidation state of iron remains unchanged.
- LiFePO_4 is prepared by mixing and grinding iron phosphate, lithium carbonate, and oxalic acid dihydrate, followed by high-temperature calcination.
- Under weakly acidic conditions, reacting LiFePO_4 with hydrogen peroxide aqueous solution can make all the lithium enter the aqueous solution, while iron and phosphorus remain in the solid phase.
- Elemental mercury, alum, and NaCl are mixed and heated in air to volatilize a compound containing mercury in the +1 oxidation state.
- In acidic aqueous solution, potassium hexathiocyanatochromate(III) is oxidized with excess I_2 , and carbon and nitrogen are only converted to ICN ; no element is produced in the reaction.

Write out the equations of the above 5 reactions (1, 3, and 5 are ionic equations), requiring the coefficients of each substance to be the simplest integer ratio. Which of the following options is correct?

- A.** All other options are incorrect
- B.** The ionic equation for reaction 1, where the sum of the product coefficients is less than the sum of the reactant coefficients.
- C.** The chemical equation for reaction 2, where the sum of the product coefficients is not a multiple of the sum of the reactant coefficients.
- D.** The ionic equation for reaction 3, the sum of coefficients of all species is a composite number.
- E.** The chemical equation for reaction 4, where the sum of the coefficients of all species is greater than 100.

F. The ionic equation for reaction 5, the sum of the coefficients of all species is greater than 150.

Answer

Correct Answer: **F**

Detailed Explanation

Reaction 1, the iron in pyrite is +2 valence, the unchanged oxidation state indicates the generation of Fe^{2+} , the +6 valence sulfur should be sulfate, and the -2 valence sulfur should form Cu_2S with the reduced copper ions. Thus, the balanced equation can be written as:



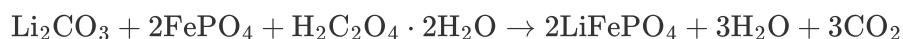
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The sum of product coefficients is 39, and the sum of reactant coefficients is 31, so option B is incorrect.

Reaction 2, the products other than lithium iron phosphate should be carbon dioxide and water. Thus, the balanced equation can be written as:



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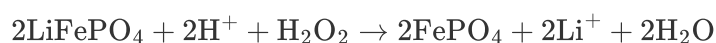
The sum of product coefficients is 8, and the sum of reactant coefficients is 4, so option C is incorrect.

Reaction 3, hydrogen peroxide acts as an oxidant to oxidize ferrous, thereby allowing lithium to enter the solution. The remaining products are iron phosphate and water. Thus, the balanced equation can be written as:

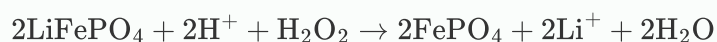
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Hydrogen peroxide acts as an oxidant to oxidize ferrous


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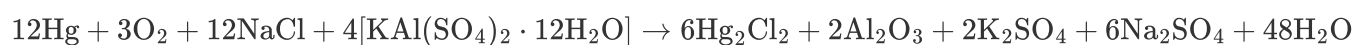
The sum of coefficients of all species is 11, which is a prime number, not a composite number, so option D is incorrect.

Reaction 4, alum is $\text{KAl}(\text{SO}_4)_2 \cdot 12\text{H}_2\text{O}$, and the gaseous compound of +1 valence mercury should be Hg_2Cl_2 . Aluminum turns into aluminum oxide, and potassium and sodium turn into sulfates. Thus, the balanced equation can be written as:

CHECKPOINT

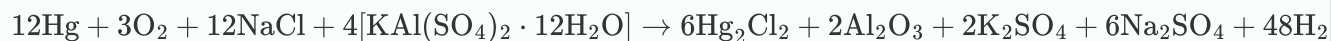
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The gaseous compound of +1 valence mercury should be Hg_2Cl_2



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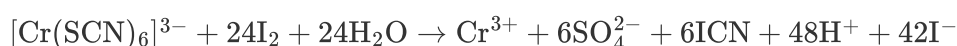
The sum of coefficients of all species is 95, so option E is incorrect.

Reaction 5, elemental iodine cannot oxidize +3 valence chromium to +6 valence, so the only thing that can be oxidized is sulfur, and the product is sulfate. The remaining products are given in the question. Thus, the balanced equation can be written as:

CHECKPOINT

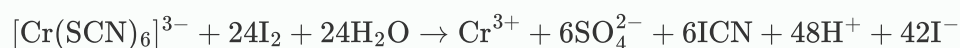
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Elemental iodine cannot oxidize +3 valence chromium to +6 valence, the only thing that can be oxidized is sulfur



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The sum of coefficients of all species is 152, so option F is correct.